
Project Proposal APS360

Mateo Tomeho

#1009985296, mateo.tomeho@mail.utoronto.ca

Link to Github: <https://github.com/mateotomeho/ActionRecognitionBasketball>

1. Introduction

This project proposal outlines my plan to develop a deep learning application for action recognition in basketball. Motivated by the rapid growth of sports analytics and my personal background as a basketball player, the goal is to build a deep learning model that can classify key basketball actions such as shooting, dribbling, passing, and defence. This will be useful and interesting because this deep neural network could be applied to automate highlight generation, analyze a player's performance and contribute to real-time broadcasting by giving some assistance with the actions occurring during a basketball game. Deep learning is an appropriate tool for this task because I can use a Convolutional Neural Network (CNN) to make image-based classification. CNNs are generally used in deep learning for computer vision and image processing, which will help to process the frames of the video to analyze the action of the players to be able to classify the basketball moves effectively.

2. Illustration

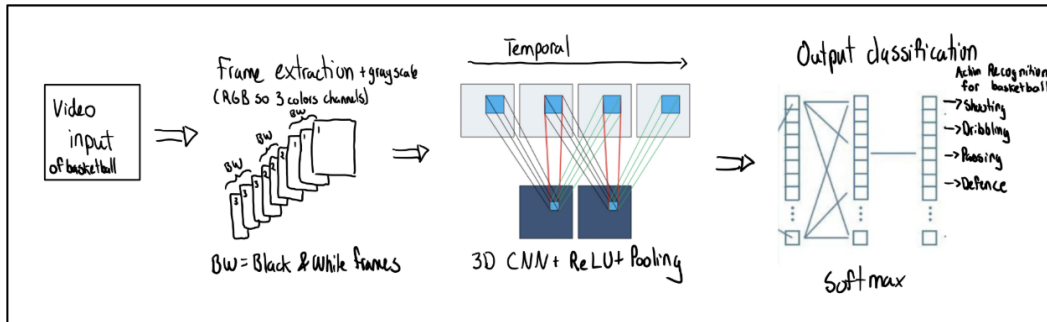


Figure 1: Illustration of the model for Action Recognition in Basketball

3. Background & Related Work

Using deep learning, many projects have been developed to recognize sports actions, including those in basketball, demonstrating the feasibility of implementing such a system in real-world contexts. The following is a brief overview of five related works on basketball action recognition.

This project [1] investigates the recognition of basketball technique actions using a three-dimensional (3D) Convolutional Neural Networks (CNNs) to automate the identification of various actions in basketball. Using 3D convolutions and Long Short-Term Memory (LSTM) networks, the model was able to get spatiotemporal relationships and temporal information of actions. The training was conducted on three public datasets: NTURGB+D, Basketball-Action-Dataset, and B3D Dataset. The proposed method achieved a 15.1% higher accuracy than the frame difference-based method and a 12.4% improvement over the optical flow method, confirming the effectiveness of the 3D CNN approach for basketball action recognition.

In this paper [2], a model was built to recognize complex actions for basketball players using the NPU RGB+D dataset.. The authors created a new feature-enhanced skeleton-based method called

LSTM-DGCN for the recognition that is based on a deep graph convolutional network (DGCN) and long short-term memory (LSTM) methods. It achieved up to 83.7% accuracy on their dataset, outperforming baseline methods and approaching state-of-the-art performance.

In this study [3], 3 modules were built to recognize basketball referee gestures (BRGR-Net). Using an Alterable Kernel Convolution (AKConv), a Coordinate Attention (CoordAtt) mechanism and a Focal Loss Function, the project was able to address the dynamic gesture, background interference and imbalances of the referee. Using a novel basketball referee gesture dataset, BRGR-Net achieved 91.2% precision in hand gesture detection.

This project [4] explores the usage of a deep neural network based on a dynamic residual attention mechanism for basketball action Recognition. The model improves upon the traditional Convolutional 3D (C3D) network by better capturing key frames and spatiotemporal features in video sequences. It will achieve an average accuracy of more than 97% for posture recognition.

In this study [5], the authors address the challenges in basketball technical action recognition by proposing a 3D convolutional neural network framework. It operates on 2 different resolution image inputs using a dataset of 1800 basketball technical action videos. The model was able to achieve an accuracy of 96.9% proving its efficiency

4. Data Processing

For this project, I will use the Space Jam dataset [6][7], which contains labelled videos of single-player basketball actions, including shooting, dribbling, defence, and passing. To prepare the data, each video will be decomposed into ordered frames to preserve the temporal evolution of the action. Then, the frames will be converted from RGB to grayscale to reduce computational complexity. The processed frames will be organized by action class, creating a clean and structured dataset suitable for training a deep neural network. The dataset will be split into 80% training, 10% validation, and 10% testing sets.

5. Architecture

Regarding the architecture, the model takes 16-frame RGB video clips (171×128 pixels) from the Space Jam dataset as input and will use a 3D Convolutional Neural Network (CNN) to capture spatiotemporal features such as the ball trajectory during a shot or the motion of a defender's stance. It will apply batch normalization to improve training stability, ReLU activations to introduce non-linearity, cross-entropy loss (CE) to measure prediction error, and Adam optimizer to update weights efficiently. After max pooling, a softmax output layer will classify the video into one of the basketball actions. This setup allows the model to learn both spatial and temporal patterns in basketball actions.

6. Baseline Model

As a baseline, we will implement a simple artificial neural network (ANN) that will take flattened frame features as input. This ANN will have 2 hidden layers and use ReLU activation. This enables comparison of 3D CNNs, which capture spatiotemporal behaviour, against ANNs that treat frames independently. Also, this model will use the cross-entropy loss and the Adam optimizer.

7. Ethical Considerations

In relation to the training data, the Space Jam dataset may not reflect the real-world diversity of basketball players, including differences in size, skin colour, playing style, and camera angles. This could introduce bias and limit generalization if applied beyond controlled settings. Regarding the model, in professional contexts, if coaches or analytics teams use the system to assist player performance, it should be used responsibly, as misclassification could negatively impact the coaching or training decisions if the system is relied on too heavily.

References

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